

ZMEINOGORSK, Russian FederationWMO: **360380**Lat: **51.1550N**Lon: **82.1722E**Elev: **1163**StdP: **14.09**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.2	-27.0	-39.1	0.6	-33.9	-32.0	0.9	-26.9	24.0	21.9	21.4	21.7	0.4	50	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	87.5	65.6	84.2	64.5	81.2	63.2	68.8	81.9	66.9	79.6	65.4	77.4	5.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.4	94.7	74.3	62.5	88.2	72.0	60.8	82.9	70.8	33.7	82.2	32.2	79.8	31.0	77.5	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.4	16.7	14.4	-36.7	93.7	8.4	3.6	-42.7	96.3	-47.6	98.4	-52.3	100.4	-58.4	103.0		
(5)			-36.8	71.7	7.9	3.7	-42.5	74.3	-47.1	76.4	-51.6	78.5	-57.3	81.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.0	5.2	9.0	22.7	41.5	54.0	64.0	66.7	63.0	52.4	40.4	23.5	11.8
	DBStd	24.36	15.45	15.00	12.29	10.47	8.76	6.87	5.56	6.48	8.34	9.56	14.46	13.86
	HDD50	6103	1388	1147	848	287	61	3	0	1	68	319	795	1186
	HDD65	10091	1853	1567	1313	707	352	97	46	114	386	761	1244	1651
	CDD50	1726	0	0	0	31	186	423	518	404	140	23	1	0
	CDD65	238	0	0	0	0	12	68	99	52	7	0	0	0
	CDH74	3143	0	0	0	17	272	798	1031	819	203	3	0	0
	CDH80	879	0	0	0	1	67	203	296	254	58	0	0	0
Wind	WSAvg	5.4	5.7	5.6	5.5	5.9	5.8	4.8	4.1	3.9	4.3	5.9	6.6	7.0
Precipitation	PrecAvg	26.1	1.7	1.6	1.5	1.9	2.5	2.2	2.9	2.6	2.1	2.8	2.6	2.1
	PrecMax	39.8	6.5	4.4	4.8	4.9	5.8	6.7	6.4	6.9	6.1	7.6	5.2	4.6
	PrecMin	19.6	0.0	0.1	0.2	0.2	0.1	0.1	0.5	0.5	0.1	0.9	0.2	0.2
	PrecStd	4.5	1.3	0.9	1.0	1.0	1.5	1.5	1.6	1.7	1.2	1.5	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	39.9	54.6	76.2	87.6	89.0	92.1	92.4	86.6	73.3	58.0	40.1
		MCWB	34.5	33.5	42.3	54.3	61.2	65.1	69.4	67.7	61.1	52.5	44.8	32.9
	2%	DB	33.9	36.0	47.2	69.9	81.0	85.2	86.6	86.5	79.9	66.4	50.8	35.4
		MCWB	30.3	31.0	38.5	51.4	58.3	64.6	67.5	65.3	58.0	49.8	41.6	31.1
	5%	DB	29.7	32.5	42.7	64.9	76.8	82.1	83.2	82.6	74.2	60.9	46.0	32.7
		MCWB	26.8	29.1	35.8	49.2	56.7	63.5	66.2	63.6	56.7	47.7	39.1	29.2
	10%	DB	26.0	29.5	39.3	59.7	72.3	78.8	80.0	78.5	69.2	55.9	41.1	29.5
		MCWB	23.4	26.3	34.4	46.9	55.1	63.1	65.0	62.2	54.4	45.6	35.9	26.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.1	34.6	43.1	55.8	63.7	69.2	73.2	70.2	63.7	54.0	46.7	34.1
		MCDB	39.7	38.3	52.9	72.8	80.9	81.6	86.4	86.5	80.9	68.5	54.2	37.2
	2%	WB	30.5	31.8	39.4	52.9	60.8	67.1	70.0	67.1	59.8	51.0	42.2	31.8
		MCDB	33.6	35.0	45.9	66.4	75.7	80.6	82.3	81.8	75.8	62.6	49.8	35.1
	5%	WB	26.9	29.2	36.8	50.4	58.6	65.6	67.8	65.2	57.6	48.7	39.3	29.4
		MCDB	29.4	32.7	41.6	62.5	71.6	78.4	79.6	78.8	71.8	59.2	45.2	32.5
	10%	WB	23.4	26.3	34.5	47.8	56.7	64.0	66.2	63.4	55.4	46.3	36.2	26.7
		MCDB	25.7	29.5	39.0	58.8	69.3	75.7	77.4	75.8	67.4	55.3	41.1	29.6
Mean Daily Temperature Range	5% DB	MDBR	19.6	20.9	19.2	21.1	24.1	22.0	20.9	24.2	23.8	18.6	15.1	16.8
		MCDBR	17.3	19.7	21.6	31.8	33.2	28.7	27.7	31.9	34.5	30.3	18.6	16.8
		MCWBR	14.7	15.9	14.7	16.8	14.6	12.1	11.8	13.4	16.0	16.4	12.2	14.0
	5% WB	MCDBR	16.1	18.8	18.3	28.4	25.9	23.5	23.0	27.1	30.9	26.9	17.2	16.2
		MCWBR	14.1	15.6	12.7	15.9	12.6	11.1	11.0	13.0	15.1	15.3	11.7	14.1
Clear-Sky Solar Irradiance	taub	0.300	0.336	0.386	0.420	0.416	0.411	0.433	0.406	0.367	0.359	0.340	0.305	
	taud	2.019	1.948	1.905	2.080	2.165	2.257	2.200	2.255	2.311	2.214	2.051	1.972	
	Ebn at Noon	213	237	248	258	266	268	259	261	259	232	193	184	
	Edh at Noon	27	38	49	46	45	41	43	39	33	30	26	24	
All-Sky Solar Radiation	RadAvg	434	786	1228	1462	1851	1999	1822	1647	1207	655	363	306	
	RadStd	31	41	113	125	148	167	133	97	106	86	38	36	